American International University–Bangladesh (AIUB)
Department of Computer Science, Faculty of Science and Technology

SE - L03 - Req. Engineering

Faculty: Mejbah Ahammad **Email:** ahammadmejbah@aiub.edu
Semester: Summer 2025-2026 **Date: June 23, 2026**

Learning Objectives

1. Define software requirements and explain their role in the Software Development Life Cycle (SDLC).
2. Differentiate between user needs and system requirements by analyzing their characteristics, objectives, and roles in software development.
3. Compare business requirements and technical requirements and explain how they contribute to successful software project implementation.
4. Distinguish between functional requirements and non-functional requirements with appropriate examples.
5. Analyze critical non-functional requirements such as performance, reliability, scalability, security, and usability in software systems.
6. Explain the phases of Requirements Engineering, including Inception, Elicitation, Elaboration, Negotiation, Specification, Validation, and Requirements Management.
7. Evaluate common challenges encountered during Requirements Engineering and discuss their impact on software quality and project success.

Learning Objective Answers

1. **Define software requirements and explain their role in the Software Development Life Cycle (SDLC).**

   Software requirements are documented descriptions of the features, functionalities, constraints, and quality attributes that a software system must satisfy. They specify what the software should do, how well it should perform, and the conditions under which it must operate. Requirements act as a bridge between stakeholder expectations and the technical implementation of a software system.

   Requirements play a fundamental role throughout the Software Development Life Cycle (SDLC). They serve as the foundation for project planning, cost estimation, resource allocation, system design, implementation, testing, deployment, and maintenance. Well-defined requirements reduce project risks, improve communication among stakeholders, and increase the likelihood of delivering software that meets user expectations.

SDLC Phase	Role of Requirements
Planning	Define project scope and objectives
Design	Guide architectural decisions
Implementation	Specify functionalities to develop
Testing	Establish validation criteria
Maintenance	Support future modifications

2. **Differentiate between user needs and system requirements by analyzing their characteristics, objectives, and roles in software development.**

   User needs represent the expectations, goals, and problems that users want the software system to address. They are generally expressed in simple language and focus on what users want to achieve. System requirements, on the other hand, provide detailed technical specifications describing how the system will satisfy those user needs.

   User needs are broad and solution-independent, whereas system requirements are specific, measurable, and testable. Understanding user needs helps identify the desired outcomes, while system requirements guide the design, development, and testing processes.

User Needs	System Requirements
High-level expectations	Detailed technical specifications
User perspective	Developer perspective
Broad and general	Specific and measurable
Focus on desired outcomes	Focus on implementation details

Example:

User Need:

"As a customer, I want to purchase products online."

System Requirement:

"The system shall provide a secure checkout process using encrypted payment transactions."

3. **Compare business requirements and technical requirements and explain how they contribute to successful software project implementation.**

   Business requirements define the organizational goals that a software system must support. They focus on why the project is being developed and what business value it should provide. Technical requirements describe the technological solutions and implementation details necessary to achieve those business goals.

   Business requirements are typically defined by stakeholders, managers, or clients, whereas technical requirements are created by developers, architects, and technical teams.

Business Requirements	Technical Requirements
Define business goals	Define technical solutions
Focus on organizational value	Focus on system implementation
Created by stakeholders	Created by technical teams
What is needed	How it will be achieved

   For example, a business requirement may state that online sales should increase by 20%. A corresponding technical requirement could specify the implementation of a secure payment gateway, recommendation engine, and mobile-responsive interface.

4. **Distinguish between functional requirements and non-functional requirements with appropriate examples.**

   Functional Requirements (FRs) describe the specific functions and services that a software system must provide. They define what the system should do and how users interact with it.

   Examples of Functional Requirements include:

   - User registration
   - User login
   - Password recovery
   - Product search
   - Order placement

   Non-Functional Requirements (NFRs) describe the quality attributes and constraints of the system. They define how well the system performs its functions.

   Examples of Non-Functional Requirements include:

   - System response time less than 2 seconds
   - 99.99% system availability
   - AES-256 encryption for sensitive data
   - Support for 10,000 concurrent users

Functional Requirements	Non-Functional Requirements
Describe system behavior	Describe quality attributes
What the system does	How well the system performs
Feature-oriented	Quality-oriented
Login, Search, Registration	Security, Reliability, Performance

5. **Analyze critical non-functional requirements such as performance, reliability, scalability, security, and usability in software systems.**

   Non-functional requirements significantly influence software quality and user satisfaction.

NFR Category	Example
Performance	Handle 1000 transactions per second
Reliability	Maintain 99.99% uptime annually
Scalability	Support 10,000 concurrent users
Security	Encrypt user data using AES-256
Usability	New users learn system within 30 minutes

   Performance ensures efficient execution of operations. Reliability ensures continuous availability. Scalability allows growth without degradation of performance. Security protects data from unauthorized access. Usability improves user experience and reduces training requirements.

6. **Explain the phases of Requirements Engineering, including Inception, Elicitation, Elaboration, Negotiation, Specification, Validation, and Requirements Management.**

   Requirements Engineering (RE) is a systematic process used to identify, analyze, document, validate, and manage software requirements. It is a cyclical and iterative process that ensures software systems satisfy stakeholder expectations and business objectives.

   The major phases of Requirements Engineering are:

   (a) **Inception**

   Inception is the initial stage of Requirements Engineering where the project idea, objectives, scope, and stakeholders are identified. This phase establishes the foundation for all future development activities.

   Activities:

   - Identifying stakeholders
   - Understanding business goals
   - Defining project scope
   - Identifying major constraints

   (a) **Elicitation**

   Elicitation involves gathering requirements from stakeholders and understanding their actual needs.

   Common elicitation techniques include:

   - Interviews
   - Surveys and Questionnaires
   - Workshops
   - Observation
   - Brainstorming Sessions

   (a) **Elaboration**

   During elaboration, gathered requirements are analyzed and refined. Ambiguities are removed and detailed specifications are developed.

   (a) **Negotiation**

   Conflicting requirements are discussed and prioritized among stakeholders to achieve agreement on project objectives and deliverables.

   (a) **Specification**

Requirements are formally documented in artifacts such as:

- Software Requirements Specification (SRS)
- Product Requirements Document (PRD)
- User Stories

The specification should be clear, complete, consistent, and verifiable.

(a) **Validation**

Validation ensures that documented requirements accurately represent stakeholder expectations and are feasible for implementation.

Validation techniques include:

- Reviews
- Inspections
- Walkthroughs
- Prototyping

(a) **Requirements Management**

Requirements Management controls requirement changes throughout the project lifecycle and ensures traceability and consistency.

Phase	Primary Objective
Inception	Identify project goals and scope
Elicitation	Gather stakeholder requirements
Elaboration	Analyze and refine requirements
Negotiation	Resolve conflicts and prioritize
Specification	Document requirements formally
Validation	Verify correctness and feasibility
Requirements Management	Control requirement changes

7. **Evaluate common challenges encountered during Requirements Engineering and discuss their impact on software quality and project success.**

Requirements Engineering is often affected by numerous challenges that can negatively impact software quality, project schedules, and costs.

Challenge	Description	Impact
Incomplete Requirements	Missing information	System failures
Changing Requirements	Frequent modifications	Delays and rework
Ambiguous Requirements	Multiple interpretations	Incorrect implementation
Communication Gap	Poor stakeholder interaction	Misaligned expectations
Lack of Domain Knowledge	Insufficient business understanding	Irrelevant solutions

One of the most common challenges is changing requirements, where business needs evolve during development. Another significant challenge is ambiguity, which can cause different team members to interpret requirements differently. Effective communication, stakeholder involvement, and proper documentation are essential for overcoming these challenges.